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Forecasting Finnish aFRR energy reserve market prices using deep learning and tree-based models

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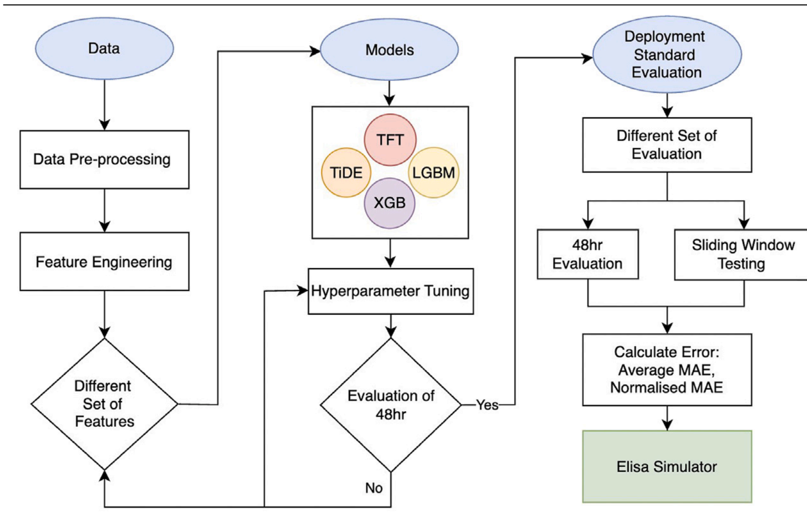
HIGHLIGHTS

- First study to forecast aFRR Down and Up marginal energy prices in Finland.
- Comparison of deep learning and tree-based models for 48-hour price prediction.
- LightGBM delivers the most consistent performance for Finnish aFRR price forecasting.
- Comprehensive feature engineering with market, weather, and time data.
- Evaluation uses three strategies to ensure robust and practical results.

GRAPHICAL ABSTRACT



aFRR energy market price forecasting pipeline

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ABSTRACT

With the increased integration of renewable energy sources, electricity systems face growing challenges due to the intermittency and volatility of weather-dependent generation. To maintain grid stability, reserve mechanisms such as the Automatic Frequency Restoration Reserve (aFRR) play a critical role by balancing supply and demand in real-time. However, the high volatility of aFRR energy prices complicates the decision-making of reserve providers and market operators. To address this challenge, this paper aims to forecast 48-hour aFRR marginal energy prices in the Finnish electricity market under both Down- and Up-regulations. We implement and compare two tree-based models, LightGBM and XGBoost, with two deep learning approaches, the Temporal Fusion Transformer and the Time-series Dense Encoder. The models are trained using historical Finnish aFRR energy prices, complemented by relevant market and weather variables, and evaluated across three cross-validation strategies. The models achieved mean absolute errors of 17.5–17.8 for Down-regulation

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and 55.7–60.1 for Up-regulation, with LightGBM delivering the most consistent performance under limited data availability. To our knowledge, this is the first study to systematically compare data-driven forecasting models for aFRR energy prices in the Finnish electricity market, providing an early-stage comparative baseline and economic relevance for a diverse array of stakeholders.

1. Introduction

Renewable energy sources (RES), including wind and solar, have the potential to reduce greenhouse gas emissions and mitigate climate change. Over the past decade, the integration of RES into electricity grids has increased worldwide. In line with this trend, the Finnish government aims to expand renewable electricity, targeting carbon neutrality by 2035 under the National Climate Act [1,2]. However, large-scale weather-dependent generation presents new operational challenges. The stochastic and intermittent pattern of RES generation can threaten grid stability and, in extreme cases, lead to large-scale blackouts, as illustrated by the 2025 Iberian blackout [3]. To manage short-term imbalances and maintain frequency stability, electricity systems rely on reserve markets that provide flexible balancing resources. Among these mechanisms, the Automatic Frequency Restoration Reserve (aFRR) is particularly important as it provides the first dynamic corrective action following frequency deviations [4].

In modern smart grids, data-driven methods are increasingly used to support monitoring and market operations [5]. Accurate forecasting under price uncertainty is particularly important for operational planning and loss minimization [6]. Within this context, forecasting aFRR demand and prices over short- and long-term horizons is crucial for system operators and market participants. Reserve providers rely on aFRR energy price forecasts as key inputs for market participation, investment decisions, and bidding optimization [7].

However, efficient bidding in the aFRR market remains challenging due to high price volatility and the complex interplay of market and external factors, such as weather conditions and system imbalances. Bidding follows a marginal pricing regime, where accepted bids are settled at the highest accepted price [8]. Therefore, reserve providers are incentivized to bid at the market's marginal price. Conversely, forecast errors can result in mismatches between actual and scheduled deliveries, which are then settled financially at potentially unstable balancing prices [9]. Market participants can thereby benefit from accurate short-term forecasting to optimize bid acceptance and revenue and minimize imbalance costs. In this study, we focus on a 48-hour forecast horizon, as it aligns with market timelines and enables the integration of forecasts into broader operational strategies, such as planning for other market variables.

Despite the central role of the aFRR in maintaining grid stability, forecasting its energy prices remains relatively unexplored. While studies have investigated aFRR price prediction in selected European markets [4,10], no prior work has addressed forecasting of aFRR energy prices in the Finnish electricity system. This gap is particularly relevant given the inherent complexity of the Nordic energy landscape [11] and Finland's market characteristics. The Finnish energy market differs from most of Europe by being weather- and interconnection-driven rather than fuel-driven, resulting in low but highly variable prices. The primary objective of this study is to develop and systematically evaluate data-driven models for short-term forecasting of Down- and Up-regulation aFRR marginal energy prices in Finland. By providing the first comparative assessment of tree-based and deep learning approaches in this market context, the study establishes a reference framework for aFRR energy price forecasting and supports informed decision-making for market participants. Specifically, this study addresses the following research questions: (i) How accurately can short-term aFRR energy prices be forecasted in the Finnish electricity market using data-driven models? (ii) How do tree-based and deep learning approaches compare in terms of predictive performance and robustness under different validation strategies? and (iii) To what extent do data availability and model complexity affect forecasting performance in this market setting?

The proposed forecasting framework uses historical Finnish aFRR energy prices, complemented by relevant market and weather variables. We implement and compare two tree-based models, LightGBM and XGBoost, and two deep learning methods, the Temporal Fusion Transformer and the Time-series Dense Encoder. Model performance is assessed using multiple error metrics across three validation strategies. The main contributions of this work are outlined below:

- An overview of the design and operation of the Finnish aFRR reserve market.
- Development and release of a cleaned and aggregated dataset combining data from Fingrid and the Finnish Meteorological Institute.
- Feature engineering and selection of key weather and market variables influencing aFRR price behavior, informed by literature, model-based feature importance, and expert input.
- Systematic comparison of tree-based and deep learning models for forecasting Down- and Up-regulation aFRR marginal energy prices.
- A robust evaluation framework using multiple performance metrics and validation techniques.
- Practical forecasting insights to support market participation and operational decision-making.

The paper is organized as follows: Section 2 reviews the Finnish aFRR market, focusing on the design and operation of the energy market. Section 3 presents an overview of prior research on forecasting methodologies for electricity and reserve market variables. Section 4 describes data sources and preprocessing steps. Section 5 outlines the general methodology, defining the forecasting objectives, feature engineering, and the modeling approaches. Section 6 details the training procedure and experimental pipeline. Section 7 presents the results evaluated under three strategies. Section 8 discusses key findings, including feature impact and model comparisons. Finally, Section 9 presents the conclusions, limitations, and future research directions.

2. Automatic frequency restoration reserve (aFRR) market

In the European power system, the grid frequency must be maintained between 49.9 and 50.1 Hz [12]. Deviations between electricity supply and demand result in frequency imbalances, which are corrected through reserve markets coordinated by Transmission System Operators (TSOs). In Finland, Fingrid fulfills this role by procuring balancing services from Balancing Service Providers (BSPs), such as power plants, storage operators, or flexible consumers [13], who adjust their output or demand in response to TSO activation signals. Activation occurs in two directions: Down and Up. During Up activation, BSPs inject energy by discharging batteries to raise the system frequency, while absorbing power during Down activation by charging batteries to lower the frequency. Fingrid organizes and procures balancing energy in four main reserve markets: the Fast Frequency Reserve (FFR), the Frequency Containment Reserve (FCR), the Automatic Frequency Restoration Reserve (aFRR), and the Manual Frequency Restoration Reserve (mFRR) [14]. Fig. 1 illustrates the activation sequence of reserve markets as the frequency drops below 50 Hz.

The aFRR is a secondary reserve mechanism designed to automatically restore the system frequency to its nominal value. It comprises a capacity and energy market. The capacity market ensures the availability of sufficient reserves and compensates BSPs for maintaining capacity,

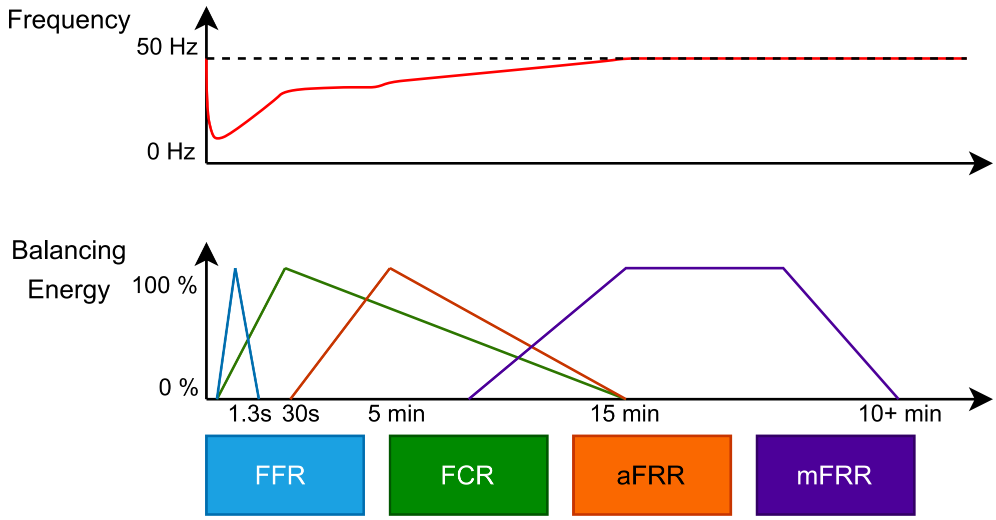

Fig. 1. Reserve activation sequence in the Finnish power system, based on [4].

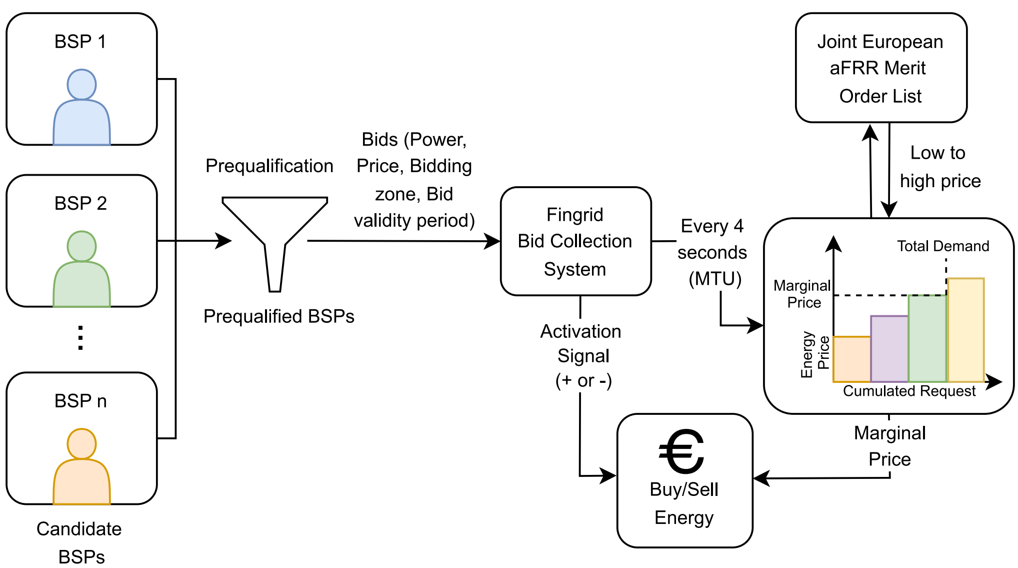

Fig. 2. Schematic representation of the Finnish aFRR energy market auction process, based on [10].

while the energy market procures balancing energy in real time and compensates BSPs for delivered energy. While aFRR was first introduced into the Nordic power system in 2013 [15], Finland launched its aFRR energy activation market on June 12, 2024, following integration into the European PICASSO platform for cross-border trading [16]. This integration enables participation in the aFRR energy market independently of the capacity market. This study examines the dynamics and price behavior of the newly established energy market.

The participation process in the Finnish aFRR energy market is summarized in Fig. 2. Following mandatory prequalification by Fingrid, eligible BSPs can submit energy bids containing the fields specified in Fig. 3. Bids are submitted through Fingrid's digital reserve trading system up to 7 days before the validity period and may be modified until 25 minutes before the start, after which they become binding [15]. For each 15-minute validity period, Fingrid submits collected bids to the joint European aFRR energy market, where they are merged into a common merit order (Up-regulation: from lowest to highest price, Down-regulation: from highest to lowest). Balancing energy is activated every 4 seconds, corresponding to the Market Time Unit (MTU), based on real-time system needs. BSPs are compensated or charged an energy fee for each MTU, determined by a uniform marginal pricing mechanism. In Up-regulation, BSPs supply energy and receive the marginal Up price, while in Down-regulation, they consume energy and are charged the marginal Down price, unless the price is negative, in which case Fingrid compensates consumption.

3. Related work

3.1. aFRR price forecasting and BESS bidding strategies

Early research on aFRR price forecasting primarily relied on statistical models and basic machine learning methods. For example, [10] compared statistical and machine learning approaches and found that statistical models provided steadier and more accurate predictions, while exogenous variables offered limited improvements. Building on this work, [17] proposed an optimization-based BESS bidding strategy for the aFRR market and concluded that profitability would require future battery cost reductions and coordinated operational strategies. These findings highlight the importance of reliable price forecasts for efficient and profitable BESS operation.

More recent studies have applied deep learning methods to reserve market forecasting. [18] evaluated feed-forward neural networks, 1-D CNNs, and LSTMs for forecasting marginal band prices in the

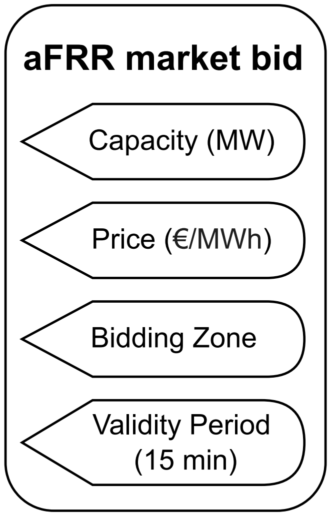

Fig. 3. Required input fields for aFRR energy bid submission.

Iberian aFRR market, finding that CNNs achieved the lowest errors and significantly outperformed previous benchmarks. Similarly, [4] compared FNN, CNN, and LSTM models for Up and Down aFRR prices in Spain, showing that FNNs performed best for Up prices, while CNNs were superior for Down prices. These results indicate that Down- and Up-regulation prices are governed by distinct market dynamics and benefit from different modeling approaches.

3.2. Day-ahead and intraday electricity price forecasting

In intraday markets, [19] applied a LASSO-based regression framework with exogenous variables and demonstrated that, contrary to earlier assumptions, such inputs improve forecasting accuracy once day-ahead prices are known. They further showed that perfect foresight of fundamental variables could reduce intraday price errors by approximately 2%.

A comprehensive review by [20] showed that inconsistent datasets and evaluation protocols hinder fair comparisons in day-ahead price forecasting. To address this, they introduced an open-access toolbox with standardized benchmarks, error metrics, and significance testing, while stressing the continued relevance of robust baseline models. Extending this work, [21] systematically evaluated machine learning models ranging from tree-based ensembles to neural networks and demonstrated that incorporating cross-border price information substantially improves forecasting accuracy, particularly during periods of extreme volatility. Recent studies have also explored advanced signal processing and ensemble methods. [22] applied Functional Data Analysis to treat daily price profiles as continuous curves, outperforming traditional time-series models such as ARIMA. Likewise, [23] showed that weighted ensemble learning significantly improves forecasting accuracy compared to single models. A significant methodological difference lies in data preprocessing. Decomposition–combination techniques with extensive filtering, such as those applied to the Italian day-ahead market [24], are effective at smoothing price dynamics. However, such approaches may be unsuitable for aFRR markets, where extreme price spikes constitute critical system signals rather than noise.

3.3. Methodological advances: Deep learning

The survey by [25] documents the widespread adoption of deep learning for time series forecasting in energy markets and provides a foundation for electricity price forecasting. While CNNs and LSTMs remain strong baselines in electricity markets [4,18], recent research has increasingly focused on complex hybrid architectures to better capture long-range dependencies. For instance, [26] proposed the EPformer, which combines BiLSTMs, Temporal Convolutional Networks, and a Transformer decoder, and employs a time–frequency loss function that explicitly penalizes errors on price peaks. Such architectures often rely on extensive preprocessing and architectural complexity. In contrast, this study examines whether specialized standalone models, such as the TFT and the computationally efficient TiDE, can deliver robust performance in the highly volatile aFRR market without hybrid model engineering.

To our knowledge, this is the first study to address aFRR energy price forecasting in the Finnish market. Prior works, such as those in Spain [4] and Iberia [18], focused on mature markets with established historical data and did not evaluate modern tree-based models alongside deep learning architectures within a unified framework. Our work fills this gap by introducing a comparative, direction-sensitive forecasting approach that explicitly models Down- and Up-regulation as distinct targets. Moreover, this paper systematically evaluates both tree-based ensembles and deep learning models. Additionally, this study incorporates lessons from previous research, such as using feature selection for various exogenous inputs, as in [18], and tailoring model choices to the behavior of each price series, as in [20], to advance forecasting for the Finnish reserve power market. This setup reflects the operational realities faced by BSPs in an emerging reserve market with limited historical data, offering both methodological rigor and practical relevance.

4. Dataset

The core dataset used in this study was obtained from the Fingrid Open Data Platform [27] and includes aFRR energy and capacity market prices, day-ahead prices, and electricity consumption data. Weather-related variables were retrieved from the Finnish Meteorological Institute [28] and were used as exogenous features. The data spans the period from June 20, 2024, to March 16, 2025. For energy Down- and Up-regulation prices, data were collected at a 15-minute granularity, while capacity prices and spot price information were available at an hourly resolution. A representative sample of the 15-minute resolution energy prices, including missing values, is illustrated in Fig. 4. The limited data span is mainly a consequence of Finland's recent entry into the PICASSO initiative, which affected market operations and data availability.

Given the objective of forecasting energy prices for the subsequent 48 h, all data were aggregated to an hourly level. This choice reflects a trade-off between volatility preservation and forecast robustness. Hourly aggregation reduces high-frequency price fluctuations and smooths isolated spikes, particularly for Up-regulation prices, but improves signal-to-noise characteristics and forecast stability for long-horizon predictions. Given the 48-hour forecasting horizon required for operational planning, modeling at a 15-minute resolution would require predicting 192 time steps ahead, which substantially increases error accumulation, especially in a newly established market with limited historical data. As a result, hourly aggregation was deemed more appropriate for reliable medium-horizon forecasting aligned with BSP planning timelines.

Temporal aggregation inevitably attenuates short-term volatility by averaging within-hour price movements. While this reduces isolated spikes, sustained price events typically driven by prolonged system imbalances remain identifiable. To quantify this effect, a comparative analysis of the September 2024 sample was performed. In the raw 15-minute series, the maximum Up-regulation price reached 556.10 EUR/MWh, which decreased to 305.46 EUR/MWh after hourly aggregation, representing a 45.1% reduction in extreme amplitude. The standard deviation, however, declined more modestly from 94.12 to 63.37, indicating that while high-frequency spikes are smoothed, the underlying market dynamics and structural variance are largely preserved.

This trade-off is acceptable for 48-hour operational planning, where reducing error accumulation across 192 time-steps takes priority over

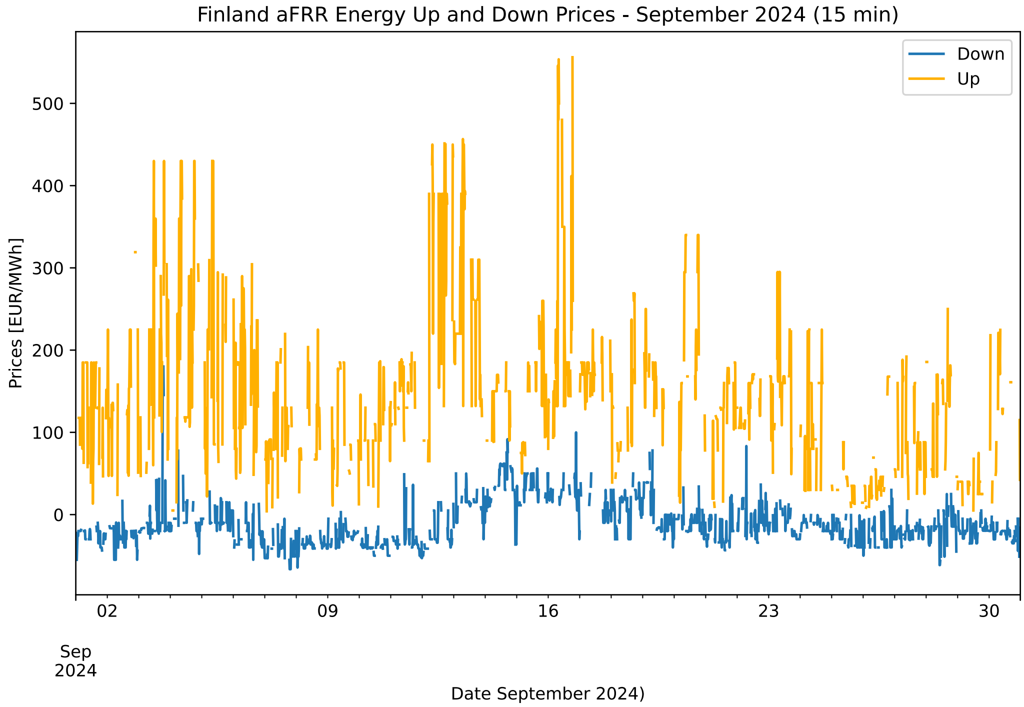

Fig. 4. Finland energy *Down* and *Up* prices for the 15-minute level dataset with missing values. The graph covers the dataset from September 1st - 30th, 2024.

capturing instantaneous 15-minute peaks. Overall, the analysis demonstrates that hourly aggregation preserves the dominant price dynamics relevant for planning-level forecasting while mitigating the forecast difficulty that would arise from modeling at a 15-minute resolution.

Prior to temporal aggregation, missing values in the 15-minute price series were imputed. Missing aFRR price observations arise from multiple mechanisms. In some cases, prices are absent due to technical issues such as API outages. In others, both Down- and Up-regulation prices are simultaneously absent, reflecting normal system conditions with no aFRR activation in either direction during the interval. Additionally, during unidirectional activations, a price is recorded only in the activated direction, while the opposite direction remains absent. Importantly, missing values do not correspond to zero prices. Empirically, zero-priced observations are extremely rare in the dataset (0.22% for Down-regulation and 0% for Up-regulation), indicating that zero is not a realistic representation of typical market conditions. Consequently, imputing missing values with zeros would introduce economically implausible price signals and bias the learning process.

Missing values were therefore imputed using a backward-looking rolling mean computed over the previous 96 observations (24 h). This window size was chosen to address a characteristic feature of the aFRR dataset: extended sequences of consecutive missing values, particularly in Up-regulation prices during periods of sustained system balance. In such cases, local imputation methods like linear interpolation would be unsuitable, as they assume a direct connection between the nearest available points and can produce unrealistic intermediate values over long gaps. The 24-hour rolling mean instead provides a stable, economically consistent baseline that reflects recent market conditions, preserving diurnal patterns without introducing artificial trends. Formally, the rolling mean imputation applied to the 15-minute price series is defined as follows:

$$P_t = \frac{1}{N} \sum_{i=0}^{N-1} P_{t-i}, \quad \text{where } N = 96 \tag{1}$$

where P_i is the observation at time i, and the window size 96 corresponds to 24 h at a 15-minute interval resolution. While this approach provides a stable baseline, the forecasting results may vary under alternative imputation strategies; future work could explore this sensitivity quantitatively. A visualization of the preprocessed hourly dataset for the aforementioned sample, showing the aFRR energy prices with imputed missing values, is presented in Fig. 5.

Outlier analysis and removal were intentionally omitted both before and after preprocessing to preserve real-world price fluctuations and avoid filtering potentially meaningful market dynamics. Removing outliers would risk suppressing important signals relevant to forecasting in a volatile market such as aFRR. After preprocessing, the final baseline dataset consisted of approximately 6456 hourly samples. Table 1 summarizes the 15 core input features used for forecasting, including market, weather, and time-based covariates.

5. Methodology

5.1. Forecasting objective

The objective is to forecast hourly Up-regulation and Down-regulation aFRR prices over a 48-hour horizon. Let P_t^{Up} and P_t^{Down} denote the Down and Up aFRR prices at hour t. Given historical data up to time t, our aim is to learn predictive functions:

$$\hat{P}^{\text{Up}}_{t+1:t+48} = f_{\text{Up}}\left(\{P^{\text{Up}}_{t-\tau}\}_{\tau=1}^{p},\ X_{t-\tau:t+q}\right), \tag{2}$$

$$\hat{P}^{\text{Down}}_{t+1:t+48} = f_{\text{Down}}\left(\{P^{\text{Down}}_{t-\tau}\}_{\tau=1}^{p},\ X_{t-\tau:t+q}\right), \tag{3}$$

where $\hat{P}^{\text{Up}}_{t+1:t+48}$ and $\hat{P}^{\text{Down}}_{t+1:t+48}$ denotes the 48-hour-ahead forecast horizon starting from time $t+1$. The parameter p is the number of past lags used as inputs. $X_{t-\tau:t+q}$ represents exogenous covariates (e.g., market features, weather data, calendar information) available from time $t-\tau$ to $t+q$, where τ is the lag index and q is the availability horizon of future known covariates. The models are trained to produce multi-step forecasts for 48 future hourly values.

5.2. Feature engineering and grouping

A comprehensive feature engineering pipeline is applied to enrich the base dataset with predictive inputs. We start from raw market data,

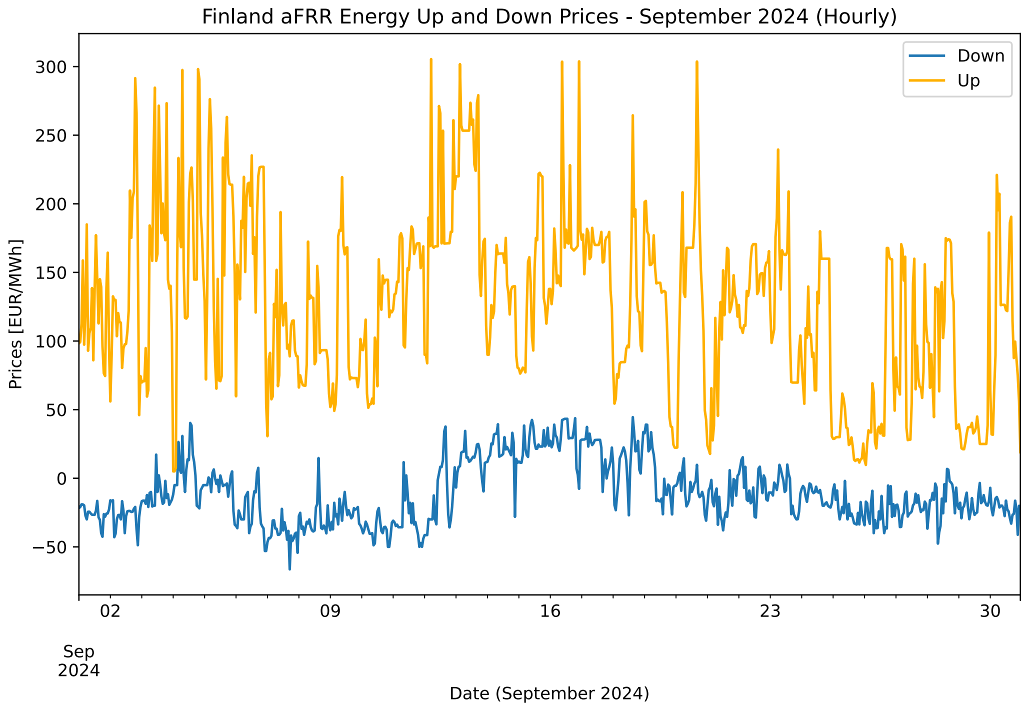

Fig. 5. Finland energy *Down* and *Up* prices for the aggregated hourly dataset with imputed missing values. The graph covers the dataset from September 1st - 30th, 2024.

Table 1
Categorized list of core features used in the final baseline dataset and their roles in forecasting.

Feature category	Feature name	Description
Market-only	Up (target/Down predictor)	aFRR energy Up-regulation price (EUR/MWh)
	Down (target/Up predictor)	aFRR Down-regulation price (EUR/MWh)
	sp	Spot price (EUR/MWh)
	Down_Cap	aFRR Down-regulation capacity price (EUR/MWh)
	Up_Cap	aFRR Up-regulation capacity price (EUR/MWh)
	electricity_consumption	Measured electricity consumption (MW)
	electricity_consumption_Finnish_networks	Total consumption across Finnish networks (MW)
	electricity_consumption_forecast	Forecasted electricity consumption (MW)
Weather-only	cloud_amount	Cloud cover index
	wind_speed	Wind speed (m/s)
	precipitation_amount	Precipitation (mm)
	pressure	Surface pressure (hPa)
	air_temperature	Ambient air temperature (°C)
	relative_humidity	Relative humidity (%)
	wind_direction	Wind direction (degrees)
Time-based	is_public_holiday	Binary indicator for public holidays

including historical aFRR prices, as well as weather data. Based on these inputs, the following feature categories are constructed:

- Lagged features: Past values of the target and key signals at multiple lags (e.g., 12, 24, 48 h) to capture autoregressive temporal dependencies.
- Rolling statistics: Rolling-window features, including mean and standard deviation, computed over relevant windows (e.g., 24h and 72 h) to capture recent variability and trends.
- Transformed weather metrics: We derived composite weather indicators such as wind chill[1] and heat index.[2] Wind direction, a circular variable, is encoded using sine and cosine components to preserve angular continuity.
- Time-based features: Cyclical time features are encoded using sine and cosine transforms [29], including hour of day, seasonal indicators (e.g., month), and binary flags for weekends and holidays. For example, hour of day is represented as

$$\sin\left(\frac{2\pi \cdot \text{hour}}{24}\right)$$

and

$$\cos\left(\frac{2\pi \cdot \text{hour}}{24}\right)$$

to capture daily cycles.
- Feature interactions and nonlinear transforms: Nonlinear transformations (e.g., logarithmic or square-root scaling) applied to skewed variables, as well as interaction terms between features with known interdependencies (e.g., demand × temperature).

[1] Calculated as $WC = 13.12 + 0.6215T - 11.37V^{0.16} + 0.3965TV^{0.16}$, where T is air temperature (° C) and V is wind speed (km/h), applied only when $T \leq 10$°C and $V > 4.8$ km/h; otherwise $WC = T$.

[2] Calculated as $HI = -8.784695 + 1.61139411T + 2.338549R - 0.14611605TR$, where T is air temperature (° C) and R is relative humidity (%), applied only when $T \geq 27$°C and $R > 40$%; otherwise $HI = T$.

Table 2
Feature groups and their composition in experimental sets.

Group	Description	Features	Example features
Group 1 (Weather)	Raw meteorological data	7	Temperature, humidity, wind speed
Group 2 (Market)	Historical market data	8	Spot price, capacity prices
Group 3 (Derived Market)	Engineered market indicators	30	Rolling averages, lagged values
Group 4 (Derived Weather)	Processed weather data	31	Rolling averages, lagged values, seasonal patterns
Group 5 (Time)	Temporal indicators	20	Day of week, holiday flags

Table 3
Composition of feature sets across feature groups.

Feature set	Weather (7)	Market (8)	Derived market (30)	Derived weather (31)	Time (20)
Set 1	3	4	11	14	8
Set 2	5	5	13	17	11
Set 3	4	6	16	19	12
Set 4	6	7	19	24	16
Set 5	7	7	23	27	19

To organize the modeling experiments, all features are categorized into five groups: (i) weather features, (ii) core market features, (iii) derived market features, (iv) derived weather features, and (v) time-based features. Different combinations of these groups were used to train model variants in order to evaluate their incremental contribution to forecast accuracy. Following the electricity price forecasting literature, calendar covariates and cyclic time encoding are well established as effective inputs in electricity price and imbalance forecasting models [22,29]. A complete list of all individual features belonging to these groups is provided in Table A.1, while the feature group combinations evaluated in the experiments are summarized in Table 2.

5.3. Feature set construction

In addition to feature grouping, we developed multiple feature sets by combining different features from all five groups. In this study, we intentionally moved beyond conventional feature elimination procedures (e.g., recursive feature elimination or exhaustive backward selection) in favor of a hybrid and iterative approach. Feature selection was guided by three principles: (i) the statistical relevance of features concerning the target variables, (ii) domain expertise provided by collaborating BSPs in electricity markets, and (iii) iterative validation of predictive performance. Our approach addresses the established drawbacks of conventional feature selection techniques in complex energy markets. Merten et al. (2020) argued in their study of the German aFRR market that correlation values and other basic statistical filters can be misleading, as they “often only consider linear relationships and neglect interdependencies between data sources”. Rigid elimination strategies can easily overlook the possibility that a feature with a low individual correlation to the target variable may still be highly predictive when interacting with other variables.

This is particularly important for the Finnish aFRR market, which experiences significant price volatility and lacks adequate historical data. As demonstrated in Fig. B.9, a basic correlation ranking and SHapley Additive exPlanations (SHAP) feature importance from our baseline LightGBM model revealed notable differences. According to SHAP, features with low correlation rankings still showed significant predictive impact. Given these findings, we relied on a hybrid approach that combined correlation checks, feature importance scoring, and expert-informed balancing of weather and market features. Through this iterative process, we created feature sets that are both operationally and statistically significant. The generated feature sets are summarized in Table 3.

5.4. Modeling approach

Our forecasting framework predicts aFRR prices 48 hours in advance using an end-to-end approach. This multi-horizon forecasting task generates a complete two-day price trajectory, enabling BSPs to submit bids 24 hours ahead. We combine two complementary modeling approaches: tree-based gradient boosting models and deep neural networks, to capture both interpretable structures and complex temporal dynamics. Each model is trained to produce a full 48-step forecast based on available input features.

We employ LightGBM [30] and XGBoost [31] due to their efficiency with medium-sized tabular datasets and ability to model nonlinear relationships with exogenous features. These ensembles are particularly robust against the noisy features typical of volatile balancing markets [32]. Complementing these, we utilize deep learning architectures to capture long-range time dependencies. We implemented the Temporal Fusion Transformer (TFT) [33] for its interpretable attention mechanisms and variable selection networks [34], and the Time-series Dense Encoder (TiDE) [35] for its computationally efficient, MLP-based global encoding capabilities [36].

All models were trained using a fixed-length historical window to forecast the next 48 h of aFRR market prices. The input window included lagged target values and rolling statistical features, providing temporal context by capturing both short-term dependencies and broader trends in the series. We distinguished between past covariates (features available up to the forecast start time, market data) and future covariates (exogenous features known over the forecast horizon, including calendar attributes and weather forecasts) [37]. All models employed a direct multi-step forecasting strategy, producing the entire 48-hour trajectory in one pass rather than iteratively predicting one step at a time.

5.5. Performance metrics

To evaluate forecast accuracy, we used five standard error metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Normalized Mean Absolute Error (nMAE), Mean Absolute Scaled Error (MASE), and Root Mean Squared Scaled Error (RMSSE). These metrics offer a comprehensive view of model performance by capturing both the scale and distribution of forecast errors. The definitions of the metrics are as follows:

- Mean Absolute Error (MAE):

$$\mathrm{MAE} = \frac{1}{N}\sum_{t=1}^{N} |F_t - A_t| \tag{4}$$

- Root Mean Squared Error (RMSE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(F_t - A_t)^2} \tag{5}$$

- Normalized Mean Absolute Error (nMAE):

$$\mathrm{nMAE} = \frac{\sum_{t=1}^{N} |F_t - A_t|}{\sum_{t=1}^{N} |A_t|} \tag{6}$$

- Mean Absolute Scaled Error (MASE):

$$\text{MASE} = \frac{\frac{1}{N}\sum_{t=1}^{N}|F_t - A_t|}{\frac{1}{N-m}\sum_{t=m+1}^{N}|A_t - A_{t-m}|} \tag{7}$$

- Root Mean Squared Scaled Error (RMSSE):

$$\text{RMSSE} = \sqrt{\frac{\frac{1}{N}\sum_{t=1}^{N}(F_t - A_t)^2}{\frac{1}{N-m}\sum_{t=m+1}^{N}(A_t - A_{t-m})^2}} \tag{8}$$

Here, F_t denotes the forecasted price, A_t represents the actual price, and N is the number of forecast steps. The nMAE is calculated using the absolute actual values in the denominator to ensure stability, even when prices are close to or below zero. The MASE and RMSSE provide scale-independent comparisons by normalizing model errors against a seasonal naïve benchmark with a seasonal period $m = 48$. A value below one indicates that the model outperforms the seasonal naïve forecast in predictive accuracy, while a value exceeding one indicates underperformance relative to this baseline.

6. Training and evaluation

6.1. Data splitting

The dataset was divided into three subsets using a 70/15/15 split: 70% for training and 15% each for validation and testing. This standard approach ensures adequate data for model learning while preserving sufficient samples for reliable evaluation.

Specifically, the training set comprised 4497 data points, the validation set 976, and the test set 960, totaling 6457 entries. While a perfect 15% split would allocate 968 data points to both the validation and test sets, a slight adjustment was made to the test set's length. To enhance compatibility with BSPs and improve forecast consistency, 8 data points were moved from the test set to the validation set. This adjustment ensured that the test set was exactly divisible into 20 segments over 48 h, allowing for more structured evaluations. The training set was used to fit the models, while the validation set guided hyperparameter tuning and early stopping. Crucially, the test set, which is chronologically available after the training and validation periods, was completely withheld during model development. All reported performance results are based solely on this unseen test set, ensuring an objective measure of each model's generalization capability.

6.2. Hyperparameter tuning

We used Optuna to optimize the hyperparameters of all key models. For tree-based models, we tuned parameters such as learning rate, number of trees, tree depth, and regularization strength by minimizing the validation MAE using the 70/15 split. For deep learning models, we optimized network architecture parameters, including the number of layers, hidden units, and dropout rates, using Optuna's Bayesian search. In each case, Optuna conducted multiple trials, where the model was trained on the training set and evaluated on the validation set. The best-performing configuration, with the lowest validation error, was selected for final evaluation on the test set.

6.3. Evaluation process

To thoroughly assess model performance, we employed three distinct evaluation methods on the test set.

Continuous Test Block Evaluation: First, the model was evaluated on the entire test set as one continuous block. In this approach, the model generates forecasts across the entire test period in sequence, calculating prediction errors over the full 15% test horizon (see Fig. 6(a)). Without being retrained or reinitialized at intermediate points, the model produces a rolling forecast. This method provides an overall view of the model's performance over a long, continuous period, treating the entire test set as one large evaluation, similar to a standard hold-out test.

Segmented 48-hour Evaluation: Second, the test set was divided into 20 non-overlapping segments, each 48 h (2 days) in length. For each segment, the model generated a 48-hour forecast using the most recent 48 h of historical data prior to that segment as context (for example, using hours $t - 48$ to t to forecast $t + 1$ to $t + 48$) (see Fig. 6(b)). Each segment was predicted independently, resetting any model state between segments, yielding 20 separate forecast windows and corresponding error estimates. By analyzing the error statistics across these multiple 48-hour segments, we assessed the variability of model performance over different periods in the test set. This protocol reveals how consistent or variable the forecast accuracy is across diverse time slices.

Sliding-Window Backtesting: We employed a sliding-window backtesting method, also known as walk-forward validation, to evaluate how well the models perform in real-world forecasting scenarios. The model was initially trained on 60 days of data and used to predict the next 48 h. For evaluation, only the last 24 h of each 48-hour forecast were considered. The final performance metric was calculated by averaging all these 24-hour segments throughout the test period. After each prediction, the window was shifted forward by 24 h (see Fig. 6(c)), and the model was fine-tuned using the newly added data. In this setup, the model generates a new 48-hour forecast every 24 h as the window advances, but it is fully retrained only every 60 days (1440 hourly steps) using all data available up to that point. This retraining helps the model avoid overfitting to old data and limits error accumulation. Between retraining intervals, the same trained model continues to produce forecasts without parameter updates. This process gradually expands the training set while maintaining chronological order, creating overlapping forecasts that reflect how the model would be used operationally, updated daily with new information, and deployed to predict the next 48 h. This approach closely mirrors a real-world scenario for BSPs, where models can be fine-tuned daily with the latest data and used to submit bids 24 h in advance. It also ensures temporal order and tests the model at multiple sequential points.

7. Results

This section presents the performance of four forecasting models—LightGBM, XGBoost, TFT, and TiDE, in predicting the aFRR Down and Up prices. We evaluated these models using the process outlined in the Evaluation section.

7.1. Continuous test block evaluation

Table 4 summarizes the performance of each model based on static evaluation using a 70/15/15 train/validation/test split. For Down-regulation, all models exhibit comparable performance, with MAE values between 17.5 and 17.8. XGBoost achieves the lowest nMAE of 0.958 and a MASE of 0.479, while LightGBM follows closely with similar scaled errors. Among deep learning models, TFT achieves the lowest RMSE (25.76), while TiDE records the highest (28.33). For Up-regulation, LightGBM delivers the strongest overall performance, although differences across models remain modest. XGBoost performs similarly in terms of MASE and RMSSE, whereas TFT shows higher MASE and RMSE due to occasional large deviations. TiDE performs weakest in this regime, with the highest MAE of 60.15. Across both regulations, all models outperform the seasonal naïve benchmark, as indicated by MASE and RMSSE values below one, confirming adequate long-horizon generalization.

Figs. 7 and 8 visually support the static evaluation results. XGBoost and LightGBM closely followed the actual data for both regulations, but they showed less adaptability to volatility. In contrast, deep learning models exhibited distinct behavioral flaws. TFT responded rapidly to local peaks but frequently overshot, whereas TiDE produced overly smoothed forecasts that lagged during sudden price shifts.

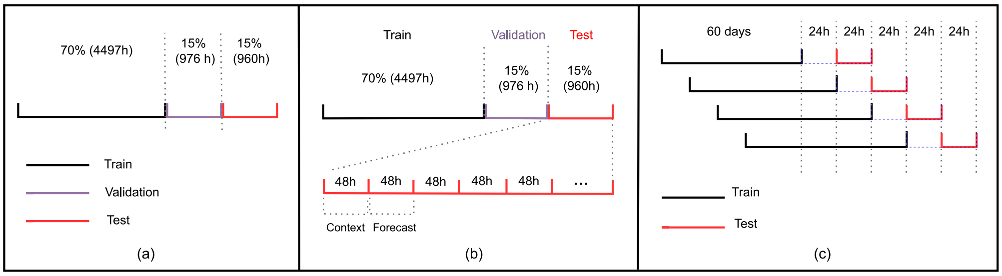

Fig. 6. Overview of the three evaluation strategies: (a) continuous test block evaluation, (b) segmented 48-hour forecasting, and (c) sliding-window backtesting approach.

Table 4
Static evaluation (70/15/15 split) of model performance on Down- and Up-regulation aFRR prices, based on a single deterministic run with a fixed random seed (42).

Model	Down-Regulation					Up-Regulation				
	MAE	nMAE	RMSE	MASE	RMSSE	MAE	nMAE	RMSE	MASE	RMSSE
LightGBM	17.58	0.963	26.32	0.482	0.569	55.74	0.603	71.17	0.498	0.553
XGBoost	17.51	0.958	26.14	0.479	0.593	57.76	0.625	69.86	0.506	0.516
TFT	17.73	0.972	25.76	0.541	0.599	57.22	0.620	79.56	0.553	0.586
TiDE	17.79	0.974	28.33	0.522	0.612	60.15	0.652	74.13	0.534	0.532

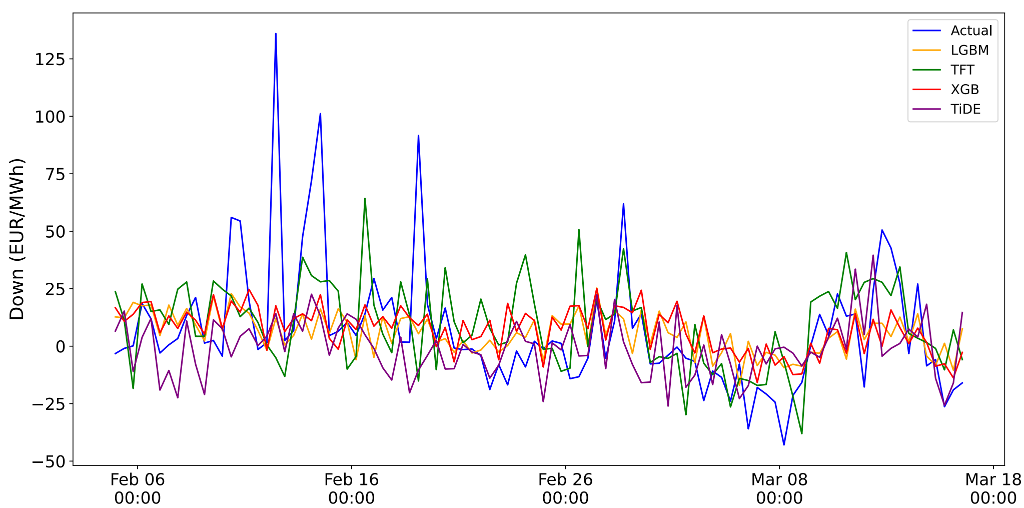

Fig. 7. Predicted vs. actual aFRR Down-regulation prices on the test set (15% of data).

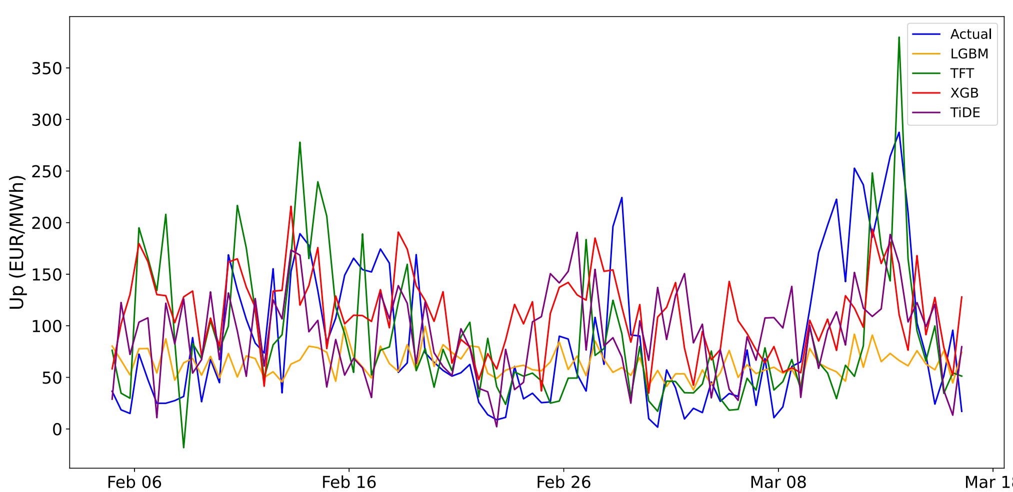

Fig. 8. Predicted vs. actual aFRR Up-regulation prices on the test set (15% of data).

Table 5
Accuracy (%) within mean absolute error thresholds for Down- and Up-regulation.

Model	Down-regulation (Accuracy %)					Up-regulation (Accuracy %)				
	±5	±10	±25	±50	±100	±5	±10	±25	±50	±100
LightGBM	22.34	42.90	79.33	93.53	98.96	7.93	15.34	34.45	61.48	81.94
XGBoost	16.81	33.61	71.82	94.26	99.06	4.28	9.50	22.44	43.11	79.96
TFT	22.08	42.29	78.44	94.69	98.85	7.71	15.31	37.40	59.48	80.52
TiDE	22.92	42.08	75.94	92.19	98.85	4.06	8.02	22.81	48.23	83.96

Table 6
Model performance across 20 non-overlapping 48-hour segments.

Model	Down-regulation (MAE)			Up-regulation (MAE)		
	Min	Avg	Max	Min	Avg	Max
LightGBM	9.41	17.64	42.47	21.74	56.71	106.11
XGBoost	9.81	17.76	42.59	32.20	54.69	102.42
TFT	9.94	19.31	41.23	22.26	73.14	107.91
TiDE	9.63	21.47	41.57	26.41	54.03	111.01

7.2. Threshold-based accuracy

To validate the evaluation of the continuous test block, we distributed the MAE with predefined threshold values of ±5, ±10, ±25, ±50, and ±100. Table 5 illustrates the percentage of predictions that fall within these defined MAE thresholds. All models demonstrated strong performance at higher threshold values (e.g., ≥93% at ±50 for Down and ≥80% at ±100 for Up). The differences between models become more pronounced at stricter thresholds. LightGBM consistently achieves the highest accuracy across both low and high thresholds, while XGBoost performs the weakest under tight error constraints. Global metrics like MAE (Table 4) suggest a marginal difference between models. The threshold-based analysis highlights an operationally relevant advantage for LightGBM. Specifically, for Down-regulation at the tightest threshold (±5), LightGBM achieves an accuracy of 22.34% compared to XGBoost's 16.81%. This constitutes a relative improvement of approximately 32% in capturing small-scale price dynamics. Although TFT performed similarly to LightGBM, it was comparatively poor in other evaluations, and TiDE did not perform as well in Up as in Down. Such differences are particularly important for market operations, where accurate prediction of small deviations can directly impact bidding and settlement outcomes.

7.3. Segmented evaluation across 48-hour windows

As shown in Table 6, we evaluated the performance of the models over 20 non-overlapping 48-hour intervals. Tree-based models outperformed the others in average MAE, while deep-learning models achieved lower maximum MAE, although the minimum MAE remained relatively similar across all models. In Down-regulation, LightGBM demonstrated consistent performance, achieving a minimum MAE of 9.41 for Down prices and an average MAE of 17.64 across all segments. XGBoost produced similar results but had a slightly higher minimum error. For the Up target class, LightGBM and TFT exhibited better performance with minimum MAEs of 21.74 and 22.26, respectively. However, TiDE stands out with an average MAE of 54.03.

7.4. Best 48-hour forecast segments

Table 7 presents the optimal 48-hour forecast window for each model, indicating the test period during which the model achieved its highest predictive accuracy. All models exhibit low MAE values for Down-regulation (9.41–9.94). Fig. C.10 shows the model's performance in the Down-regulation case. LightGBM demonstrated better short-term responsiveness compared to the seasonal naïve benchmark. Alternatively, XGBoost had less stable adaptation to the 48-hour forecast period, performing well in raw error metrics but exhibiting higher scaled errors. TFT proved competitive (nMAE 0.512 for Up-regulation) and effectively captured volatile points in Fig. C.11. However, both deep learning models struggled with scaled error metrics, particularly TiDE in both Down- and Up-regulation, where the naïve model outperformed them.

7.5. Sliding window evaluation

The results of the sliding window evaluation are presented in Table 8. During this evaluation, we periodically train models for continuous forecasting that must adapt to changing data patterns. In Down-regulation, TiDE demonstrated maximum stability by achieving the lowest MAE (20.66) and minimal drift. Other models exhibited MASE and RMSSE values exceeding 1.0, indicating lower robustness than the seasonal naïve benchmark. This behavior can be partly attributed to the narrow price range of Down-regulation closer to zero, where small absolute deviations are magnified in scaled error metrics, and frequent retraining may amplify reactions to short-term volatility. Conversely, in Up-regulation, characterized by higher, strictly positive price levels, all models consistently outperformed the seasonal naïve benchmark, as reflected in MASE and RMSSE values below 1.0. Figs. C.12 and C.13 visualize these predictive behaviors over time. Predicting in a sliding window is particularly challenging due to high signal variability and limited training data at the start of the evaluation window.

7.6. Evaluation reasoning and insights

The practical goals of using multiple model evaluation strategies include selecting the best-performing model class and testing performance across different data availability and optimization timelines. Furthermore, these evaluations assess long-term changes in performance, e.g., due to data drift, changes in the data-generating process, and structural breaks.

The continuous test block with test, train, and validation split evaluation allowed us to conclude that gradient boosting frameworks are one of the best choices for predicting price movements in aFRR energy markets. This was also supported by findings using a segmented 48-hour window evaluation. This specific window length was selected in order to comply with the input requirements of the BSP's optimization algorithm. Using fixed two-day windows allowed us to assess predictors' performance for daily planning under normal operational conditions. Finally, the sliding windows approach and continuous retraining were tested to see how the models handle possible data drift, significant changes in market regulation, and unexpected events (e.g., changes in aFRR prices due to blackouts, unscheduled maintenance breaks of large generating facilities). This approach is the closest to the practical deployment of forecasting algorithms used at the BSP. It also allowed us to obtain ballpark execution time and efficiency figures for the algorithms. Table D.2 summarizes the analytical objectives and the specific insights derived from each evaluation.

7.7. Comparison with previous studies

To provide context for our results, we compare them with recent studies on aFRR price forecasting in other European markets, specifically the Spanish aFRR study by [4] and the German aFRR study by [10]. The Spanish study used deep learning techniques, employing the previous 96 quarter-hour prices as predictors for the next day, and reported scaled errors based on a persistence baseline. [4] incorporated results

Table 7
Best 48-hour split evaluation results for each model.

Model	Down-regulation					Up-regulation				
	MAE	nMAE	RMSE	MASE	RMSSE	MAE	nMAE	RMSE	MASE	RMSSE
LightGBM	9.41	1.37	12.02	0.399	0.315	21.74	0.566	26.19	0.365	0.259
XGBoost	9.81	1.43	11.78	1.048	1.130	32.20	0.544	43.32	0.732	0.689
TFT	9.94	1.11	12.59	0.782	0.791	22.26	0.512	27.65	1.666	1.565
TiDE	9.63	0.886	12.18	1.039	0.944	26.41	0.521	33.40	1.135	1.123

Table 8
Sliding window evaluation results for forecasting Down- and Up-regulation prices.

Model	Down-regulation					Up-regulation				
	MAE	nMAE	RMSE	MASE	RMSSE	MAE	nMAE	RMSE	MASE	RMSSE
LightGBM	33.53	1.597	40.41	1.631	1.412	61.06	0.657	81.99	0.833	0.823
XGBoost	30.52	1.453	37.49	1.356	1.249	60.60	0.652	82.19	0.842	0.844
TFT	32.52	1.549	39.30	1.821	1.561	67.57	0.727	92.99	0.949	0.960
TiDE	20.66	1.075	27.17	0.997	0.946	61.83	0.819	83.81	0.843	0.841

Table 9
Comparison of forecasting performance with prior aFRR price forecasting studies.

Reference	Down-regulation		Up-regulation	
	MASE	RMSSE	MASE	RMSSE
This work	0.482–0.541	0.569–0.612	0.498–0.553	0.516–0.586
[4]	0.7180	0.7594	0.6895	0.7202
[10]	0.7833	0.8013	0.9118	0.9051

from [10] and normalized the original MAE and RMSE values using the same persistence model for consistent cross-market comparison. Although our baseline differs by employing a seasonal-naïve model with a 48-hour periodicity, we ensure consistency by basing our comparison on the normalized MASE and RMSSE values from the Spanish study. The MASE values range from 0.482 to 0.541 for Down-regulation prices and from 0.498 to 0.553 for Up-regulation prices, corresponding to the range observed across the four evaluated models. Our models demonstrate significantly lower errors. Despite differences in temporal resolution, market structure, and baseline definitions, our experimental setup indicates that contemporary architectures like LightGBM and TiDE can achieve competitive accuracy, and in some cases, even better accuracy (see Table 9).

This comparison should be approached with caution. The studies use different data granularities and operate in different market contexts. Our dataset includes hourly aFRR prices, while the German market employs 4-hour trading blocks, and the Spanish study used 15-minute intervals. These variations can complicate forecasting and increase potential errors. However, we can still compare forecasts across markets using metrics like MASE and RMSSE, suggesting that our models are reliable, especially in a European context.

8. Discussion

In this study, we benchmarked deep learning architectures against tree-based ensembles in the Finnish aFRR energy market. To ensure a robust evaluation, strict temporal separation was enforced between training, validation, and test datasets to prevent data leakage. Techniques such as early stopping, dropout, and regularization were employed to mitigate overfitting and ensure that the reported results reflect true generalization rather than memorization of historical patterns. The evaluation strategies adopted in this study are motivated by a research perspective aimed at analyzing market behavior through multiple analytical lenses. They do not imply that BSPs currently follow or are required to follow these specific evaluation practices.

After evaluating various strategies, LightGBM consistently demonstrated robust performance. XGBoost performed comparably in many evaluations, but the threshold-based accuracy analysis highlights why LightGBM is a better competitor. Among deep learning models, TiDE excelled in the sliding window evaluation for Down-regulation, whereas TFT offered high short-term accuracy but proved sensitive to volatility. Our models demonstrate competitive performance relative to recent European aFRR studies. Since direct comparison is complicated, our MASE values of 0.482–0.541 (Down) and 0.498–0.553 (Up) substantially outperform both the Spanish market [4] (MASE: 0.72/0.69) and the German market [10] (MASE: 0.78/0.91). This higher performance may reflect both methodological improvements and structural differences in the Finnish market.

Crucially, aggregate metrics (MAE, RMSE) alone are insufficient to characterize operational usefulness in the aFRR context. In Down-regulation, where prices often hover near zero, scaled metrics (nMAE, MASE) amplify small absolute deviations, distorting comparative assessments. From an operational perspective, value lies less in minimizing global average error and more in maintaining predictions within economically tolerable bounds. Our threshold-based analysis demonstrates that evaluating performance within these decision-relevant margins reveals operational distinctions that standard aggregate metrics obscure. Additionally, in a pay-as-bid market, deviations beyond narrow tolerance bands lead directly to mispriced bids and lost revenue. Consequently, LightGBM's superiority in this study should be interpreted not as a marginal reduction in statistical error, but as higher consistency in producing economically viable price estimates. This viability is reinforced by its unique stability in all evaluations, effectively avoiding the volatility spikes observed in deep learning models. Furthermore, its computational efficiency allows for frequent retraining to capture rapid market shifts, a critical deployment requirement that heavier architectures cannot easily satisfy.

The configuration of the forecast horizon further reflects these operational realities. A 48-hour context window was selected to align with the Finnish aFRR market structure, in which BSPs submit bids approximately 40 hours in advance. While longer historical windows were considered, aFRR price formation is driven by rapidly evolving system imbalances. Extending the context window risks introducing obsolete information that no longer correlates with the current state of the grid. The chosen window, therefore, optimizes the trade-off between data freshness and sufficient coverage of the day-ahead planning horizon. This necessity to capture rapid, short-term imbalances is most critical for Up-regulation prices, which proved significantly more challenging to forecast due to their higher volatility compared to Down-regulation.

To interpret model behavior under these volatile conditions, we employed SHAP analysis. It is important to note that feature importance indicates predictive leverage, not necessarily physical causality. aFRR prices emerge from proprietary BSP bidding strategies, TSO activation constraints, and portfolio-level decisions that are not publicly observable. Nevertheless, the observed divergence in feature importance is economically plausible. Up-regulation prices tend to be scarcity-driven and sensitive to system stress, whereas Down-regulation prices are influenced by surplus conditions like high wind generation. These distinctions align with known market behavior, validating that the models are learning relevant economic signals rather than noise.

Several contextual limitations must be considered when interpreting the relative performance of deep learning models. Most notably, the Finnish aFRR market has been operational only since mid-2024, resulting in a dataset of approximately 6456 hourly observations. This sample size is limited for high-capacity architectures such as the TFT, which typically require substantially larger datasets to effectively generalize. In such small-sample regimes, the risk of overfitting increases, and complex models may exhibit higher variance and reduced stability compared to simpler alternatives. This limitation helps explain why tree-based models performed more consistently and were more computationally efficient. No formal learning curve or ablation analysis was conducted, as further reducing the already constrained training set would compromise statistical validity. As the aFRR market matures and longer time series become available, revisiting deep learning scalability and learning-curve behavior represents a natural extension of this work.

To address limited data and capture external and temporal signals, we developed an extended feature engineering approach that incorporated over 90 raw and derived features. An exhaustive search for all possible combinations (i.e., 2^{95}) is computationally infeasible. Although we relied on feature importance scoring, domain knowledge, previous studies, and iterative validation, there remains a risk of overlooking relevant variables. The complexity and computational demands of models are crucial to practical applicability, underscoring the operational advantages of the lightweight tree-based approaches identified in this study.

Finally, upon our request, partner BSP simulated its revenues based on an optimization that uses LightGBM's aFRR energy price forecast. Later, it compared this to a naïve benchmark forecast, keeping everything else equal for a sample day trade. The simulation indicated a substantial reduction in imbalance-related sanctions (approximately 23%) and a modest increase in total simulated revenues (around 1%). The result is intuitive: a more accurate energy price prediction helps estimate the opportunity costs of actions taken in the market. It is crucial for market operations as it allows finding the optimal timing for various market activities in order to maximize revenues and avoid sanctions.

9. Conclusions and future work

This study presents a comprehensive analysis of deep learning and tree-based models to forecast the aFRR energy prices in the Finnish energy grid system. By framing the forecasting task through multiple complementary evaluation strategies, the study examined not only average predictive accuracy but also robustness, adaptability, and decision-relevant precision under realistic market constraints.

The results demonstrate that different testing regimes reveal distinct aspects of forecast behavior. Within this framework, LightGBM consistently performed well, while XGBoost's performance was slightly worse but still comparable. TiDE demonstrated adaptability when retrained over time, while TFT showed potential but would benefit from a larger training dataset. The best models achieved mean absolute errors ranging from 17.5 to 17.8 for Down-regulation and 55 to 60 for Up-regulation. The average predicted Down-regulation price was 8.67 €/MWh, and the average Up-regulation price was 64.70 €/MWh. Up-regulation prices, while more volatile, resulted in smaller normalized prediction errors due to their higher price level.

A key methodological insight is that conventional aggregate error metrics alone are insufficient to assess practical forecasting value in aFRR markets. This perspective emphasizes consistency and decision-relevant precision over marginal reductions in global error statistics. The study is inherently constrained by the limited historical depth of the Finnish aFRR market. Accordingly, this work should be viewed as an early-stage benchmark rather than a definitive ranking of model classes.

Future research should prioritize reassessing the scalability of deep learning models as historical datasets expand, while also developing hybrid architectures that merge LightGBM's consistency with TiDE's drift resistance. Furthermore, this framework should be extended to other Nordic and European aFRR markets. This cross-market approach would also validate the applicability of our findings across different market structures and bidding protocols.

CRediT authorship contribution statement

Abu Saleh: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Aruzhan Amangeldina:** Writing – original draft, Visualization, Software, Methodology, Investigation. **Tesfay W. Tesfay:** Writing – original draft, Visualization, Software, Methodology, Data curation. **Sébastien Lafond:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Hergys Rexha:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization. **Olena Izhak:** Writing – review & editing, Validation, Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Complete feature list

See Table A.1.

Appendix B. Feature construction

A key justification for our hybrid feature selection approach is that the model itself shows a clear difference between simple linear correlation and true predictive importance. Although a correlation analysis is a good place to start, it can be deceptive because it ignores feature interactions and non-linear relationships, which are crucial in the intricate energy market. For example, the SHAP analysis ranks `dayofyear` as the second most impactful feature, despite its weak correlation of only 0.235, suggesting a strong non-linear, seasonal influence that correlation is unable to capture. Similarly, `Down_Cap` is the third most important feature according to SHAP, despite having a negligible correlation of 0.084. This suggests that the way it interacts with other variables is what gives it its predictive power. Even `wind_direction` aids in prediction, presumably through conditional effects that depend on other meteorological factors, as evidenced by its nearly zero correlation of 0.004. These examples show how some of the model's most useful predictors would have been mistakenly eliminated by a strict, correlation-based feature elimination.

Appendix C. Additional evaluation illustrations

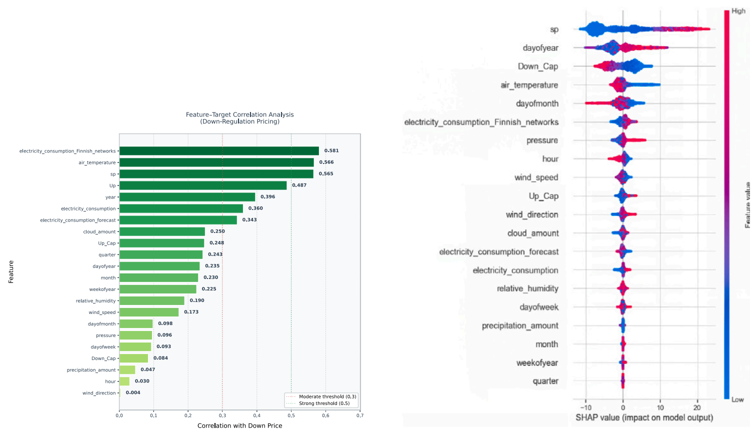
See Figs. C.10–C.13.

Appendix D. Evaluation insights

See Table D.2.

Table A.1
Complete list of all features used in the forecasting model, including the 15 baseline input features as well as all additional derived, engineered, and time-based features.

Feature category	Feature name	Description
Market-only	Up (target/Down predictor)	aFRR Up-regulation price (EUR/MWh)
	Down (target/up predictor)	aFRR Down-regulation price (EUR/MWh)
	sp	Spot price (EUR/MWh)
	Down_Cap	aFRR Down-regulation capacity price (EUR/MWh)
	Up_Cap	aFRR Up-regulation capacity price (EUR/MWh)
	electricity_consumption	Measured electricity consumption (MW)
	electricity_consumption_Finnish_networks	Total consumption across Finnish networks (MW)
	electricity_consumption_forecast	Forecasted electricity consumption (MW)
Weather-only	cloud_amount	Cloud cover index
	wind_speed	Wind speed (m/s)
	precipitation_amount	Precipitation amount (mm)
	pressure	Surface pressure (hPa)
	air_temperature	Ambient air temperature (°C)
	relative_humidity	Relative humidity (%)
	wind_direction	Wind direction (degrees)
Derived market	Down_lag_12, Down_lag_24, Down_lag_36, Down_lag_48	Down price value from 12, 24, 36, and 48 h ago
	Up_lag_12, Up_lag_24, Up_lag_36, Up_lag_48	Up price value from 12, 24, 36, and 48 h ago
	sp_lag_12, sp_lag_24, sp_lag_36, sp_lag_48	Spot price value from 12, 24, 36, and 48 h ago
	electricity_consumption_lag_24, electricity_consumption_lag_48	Electricity consumption value from 24 and 48 h ago
	Down_rolling_mean_12, Down_rolling_mean_24	Mean of Down price over the past 12 and 24 h
	Up_rolling_mean_12, Up_rolling_mean_24	Mean of Up price over the past 12 and 24 h
	Down_rolling_std_12, Down_rolling_std_24	Standard deviation of Down price over the past 12 and 24 h
	Up_rolling_std_12, Up_rolling_std_24	Standard deviation of Up price over the past 12 and 24 h
	sp_mean_24, sp_mean_48	Mean of Spot price over the past 24 and 48 h
	sp_std_12, sp_std_24	Standard deviation of Spot price over the past 12 and 24 h
	electricity_consumption_rolling_mean_24	Mean of electricity consumption over the past 24 h
	electricity_consumption_Finnish_networks_rolling_mean_24	Mean of Finnish networks electricity consumption over the past 24 h
	log_Down	Log-transformed Down price (variance stabilization)
	log_Up	Log-transformed Up price (variance stabilization)
Derived weather	lag_temp_3h, lag_temp_6h, lag_temp_9h	Air temperature value from 3, 6, and 9 h ago
	lag_pressure_6h	Surface pressure value from 6 h ago
	rolling_temp_6h, rolling_temp_12h	Mean of air temperature over the past 6 and 12 h
	rolling_humidity_6h	Mean of relative humidity over the past 6 h
	wind_direction_sin, wind_direction_cos	Circular encoding of wind direction
	wind_chill	Perceived temperature accounting for wind effect
	heat_index	Perceived temperature accounting for humidity effect
	dew_point	Dew point temperature
	wind_power	Proxy for wind energy potential (wind speed squared)
	storm_index	Combined indicator of wind speed and precipitation
	humidity_x_wind	Interaction term: humidity and wind speed
	wind_speed_x_hour	Interaction term: wind speed and hour of day
	relative_humidity_x_air_temp	Interaction term: humidity and air temperature
	wind_speed_x_dayofweek	Interaction term: wind speed and day of week
	precipitation_x_hour	Interaction term: precipitation and hour
	temp_x_hour, temp_x_dayofweek, temp_x_month	Interaction terms: air temperature with hour, day of week, and month
	temp_diff_1h, temp_diff_24h	Temperature change over 1 and 24 h
	pressure_change_6h, pressure_change_12h	Pressure change over 6 and 12 h
	log_wind_speed	Log-transformed wind speed
	sqrt_precipitation	Square-root transformed precipitation
	is_high_wind	Binary indicator for high wind speed
	is_heavy_rain	Binary indicator for heavy rainfall
	is_heat_wave	Binary indicator for heatwave conditions
Time-based	hour, dayofweek, month, year, dayofyear, dayofmonth, weekofyear, quarter	Calendar-based time features
	hour_sin, hour_cos	Cyclical encoding of hour of day
	dayofweek_sin, dayofweek_cos	Cyclical encoding of day of week
	month_sin, month_cos	Cyclical encoding of month
	week_sin, week_cos	Cyclical encoding of week of year
	is_weekend	Binary indicator for weekend
	is_summer	Binary indicator for summer season
	is_winter	Binary indicator for winter season
	is_public_holiday	Binary indicator for public holidays



(a) Feature–target correlation (Down price)

(b) SHAP feature importance (LightGBM, Down price)

Fig. B.9. Comparison between linear correlation and SHAP-based feature importance for Down-regulation price forecasting. The correlation plot (left) shows linear relationships, while the SHAP plot (right) captures non-linear and interaction effects.

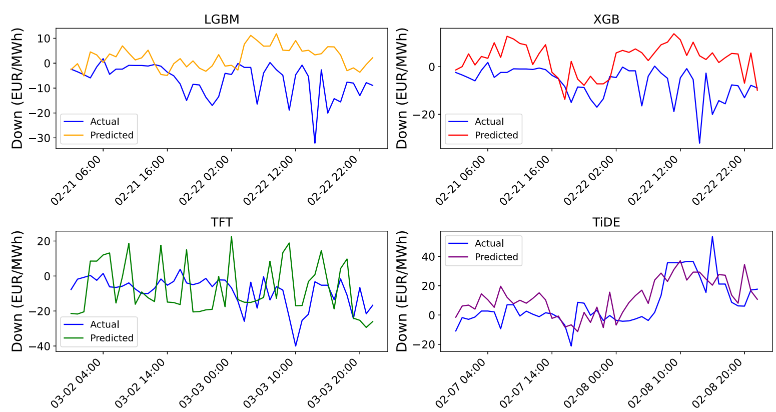


Fig. C.10. Predicted vs actual aFRR Down-regulation prices for the best 48-hour window.

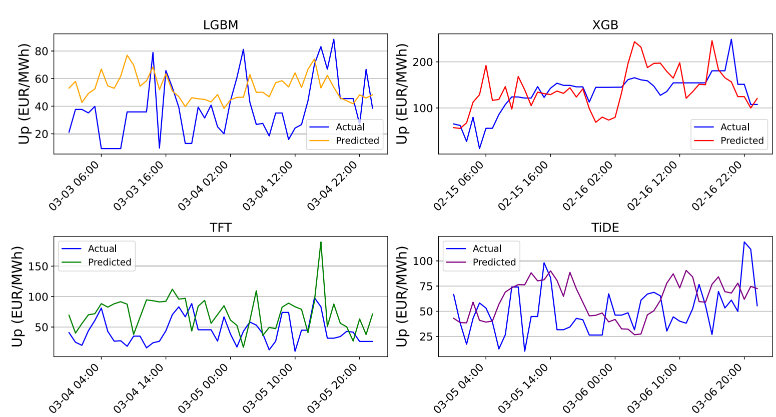


Fig. C.11. Predicted vs actual aFRR Up-regulation prices for the best 48-hour window.

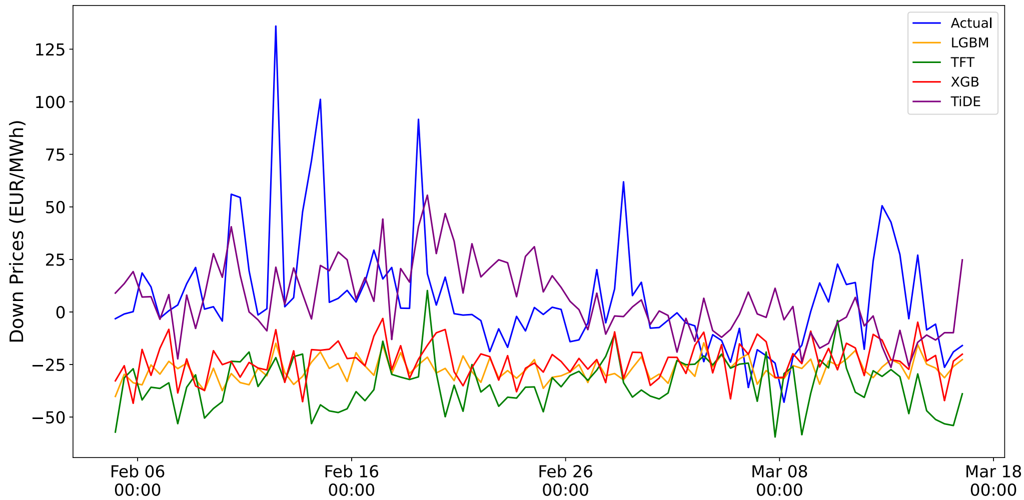


Fig. C.12. Predicted vs. actual aFRR Down-regulation prices using sliding window evaluation, showing the reference window (Feb 5–Mar 16, 2025) characterized by high price variability.

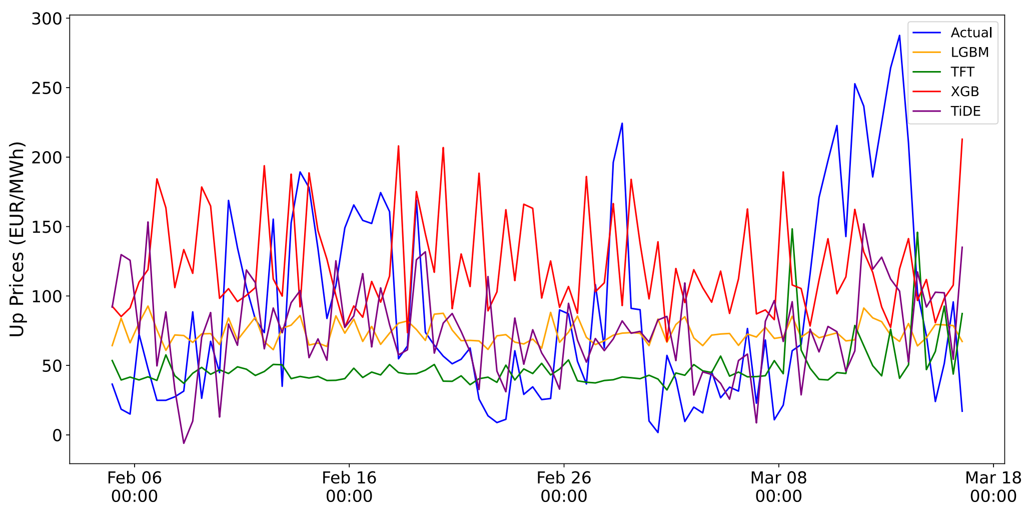


Fig. C.13. Predicted vs. actual aFRR Up-regulation prices using sliding window evaluation, showing the reference window (Feb 5–Mar 16, 2025) characterized by high price variability.

Table D.2
Synthesis of evaluation strategies, operational relevance, and primary findings.

Evaluation strategy	Operational context	Key findings
Continuous Test Block (70/15/15 split, single 960-hour forecast)	Baseline generalization without retraining. Tests long-horizon forecasting capability and overall model accuracy.	LightGBM achieves the best balance across Up/Down regulation (lowest MASE). While TiDE is competitive in specific metrics, LightGBM and XGBoost offer the most consistent “out-of-the-box” fit for long horizons.
Segmented 48h Windows (20 non-overlapping blocks, independent forecasts)	Day-ahead bidding reliability across diverse market conditions. Assesses forecast stability and variance under regime shifts.	LightGBM achieves the lowest MAE in the “Best 48h” scenarios (Down: 9.41, Up: 21.74). LightGBM is the safest choice for daily planning, as it minimizes error during standard market conditions better than deep learning alternatives.
Sliding Window (60-day training, 48h forecast, 24h step, periodic retraining)	Production deployment with concept drift. Simulates a real-world lifecycle where models adapt to evolving patterns through periodic retraining.	TiDE is highly stable in Down-regulation (lowest MAE 20.66), but LightGBM outperforms it significantly in Up-regulation Threshold Accuracy (7.93% vs 4.06%). TiDE is a “specialist” for stable trends, while LightGBM is a “generalist” that handles the volatility of both Up and Down markets effectively.

Data availability

The supporting dataset for this study has been published and is publicly available in the Zenodo repository at https://doi.org/10.5281/zenodo.17494556.

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